

Notes: Dessaint, Foucault, Frésard, and Matray (RFS, 2019)

Noisy Stock Prices and Corporate Investment

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1 Big picture

- **Question.** Do nonfundamental movements in stock prices affect real corporate investment, even when firms are not constrained and managers are not trying to cater to mispricing?
- **Core idea.** Stock prices are signals. Managers may use their own stock price and peers' stock prices to infer investment opportunities. If they cannot perfectly separate information from noise, then noisy prices become *faulty informants*.
- **Identification idea.** Instead of using the noise in a firm's own stock price, the paper uses nonfundamental shocks to product-market peers' stock prices. This is cleaner because financing and agency channels tied to the firm's own stock price should not directly operate through peers' prices.
- **Noise measure.** The paper uses hypothetical sales by mutual funds experiencing large outflows. These sales generate fire-sale price pressure and are unlikely to reflect fundamentals of the affected firms.
- **Setting.** U.S. nonfinancial, nonutility firms from 1996–2011. Product-market peers are identified with Hoberg–Phillips TNIC industries.
- **Main result.** A one-standard-deviation decrease in the nonfundamental component of peers' stock prices reduces a firm's investment by about 1.5 percentage points, or 4.3% relative to the sample mean investment level.
- **Aggregate magnitude.** The authors estimate an aggregate loss in investment of about \$29 billion per year and shareholder opportunity costs of roughly \$0.9–\$3.7 billion per year.
- **Dynamic pattern.** The real investment effect of noise is transient. Investment falls after the nonfundamental peer-price shock and then reverses. Fundamental peer-price information has a more persistent investment effect.
- **Cross-sectional pattern.** The effect is stronger when managers are less likely to identify the noise and when peer prices are more informative about the firm's own fundamentals.
- **Omitted-variable check.** In conglomerates, division-level tests with firm-year fixed effects show that noise in the stock prices of a division's peers affects capital allocation across divisions within the same firm-year.
- **Alternative channels.** The paper rules out financing constraints, pressure from product markets or managerial incentives, investment mimicking, and correlated own-price noise as the main explanations.
- **Takeaway.** Even if managers maximize long-run value and are financially unconstrained, stock-market noise can affect the real economy because managers learn from prices that are only imperfectly informative.

2 Data and measurement

2.1 Sample construction

- The baseline sample covers U.S. firms from 1996 to 2011.
- Financial firms and utilities are excluded.
- Firm-years with negative sales, sales below \$5 million, or missing information on assets, capital expenditures, PPE, or end-of-year stock prices are excluded.
- The main sample has 45,275 firm-year observations.
- Variables are winsorized at the 1% level in each tail.

Interpretation: The sample is designed to study ordinary corporate investment decisions rather than regulated firms, financial firms, or distressed observations with problematic accounting values.

2.2 Investment outcome

The main investment variable is

$$I_{i,t} = \frac{\text{Capex}_{i,t}}{\text{PPE}_{i,t-1}}, \quad (1)$$

where capital expenditures are scaled by lagged property, plant, and equipment.

Interpretation: This is a standard investment measure. The baseline coefficient can be read as the change in investment intensity for a one-standard-deviation change in a price-noise variable.

2.3 Product-market peers

- Peers are defined using the Text-Based Network Industry Classification (TNIC) developed by Hoberg and Phillips.
- For each firm-year, the paper constructs averages of peer variables excluding the focal firm.
- Main peer aggregation is equal-weighted, but results also use similarity-weighted averages, medians, five closest peers, and aggregate peer portfolios.

Interpretation: The peer definition is central. The paper wants stock prices of firms whose fundamentals are correlated with the focal firm, because these are the prices managers might rationally watch.

2.4 Nonfundamental price-pressure measure

The nonfundamental component of a stock price is proxied using mutual fund hypothetical sales:

- A mutual fund is considered shocked when it experiences large outflows, above 5% of its assets.
- Hypothetical sales assume the fund proportionally liquidates its existing holdings to keep portfolio weights unchanged.
- For each stock, these hypothetical sales are scaled by trading volume.
- The variable is denoted $\text{MFHS}_{i,t}$ for the firm and $\text{MFHS}_{-i,t}$ for the peer portfolio.
- MFHS takes negative values; more negative values indicate stronger fire-sale pressure.

Interpretation: Because hypothetical sales come from investor redemptions rather than fund managers' stock-picking decisions, they are plausibly nonfundamental demand shocks.

2.5 Decomposing stock prices

The paper decomposes peers' Tobin's q into a nonfundamental component and a residual component. Conceptually,

$$Q_{-i,t} = \lambda_i + \delta_t + \phi \text{MFHS}_{-i,t} + \hat{\nu}_{-i,t}. \quad (2)$$

The fitted component based on $\text{MFHS}_{-i,t}$ is the observable nonfundamental component. The residual is denoted

$$Q_{-i,t}^* \quad (3)$$

and is used as the component of peers' stock price unexplained by mutual-fund hypothetical sales. The same decomposition is applied to the firm's own $Q_{i,t}$.

Interpretation: The paper does not claim that Q^* is perfectly fundamental. It is the component orthogonal to the specific fire-sale noise used in the empirical design.

2.6 Controls

The baseline controls include lagged values of:

- the firm's own nonfundamental price component $\text{MFHS}_{i,t-1}$;
- the firm's own residual stock-price component $Q_{i,t-1}^*$;
- peers' residual stock-price component $Q_{-i,t-1}^*$;
- firm and peer cash flow;
- firm and peer size;
- peer investment.

Firm fixed effects and year fixed effects are always included in the baseline specification.

Interpretation: The coefficient of interest is therefore not just the response to peer prices. It is the response to the *nonfundamental* part of peer prices, controlling for own price information and peer fundamentals.

3 Models and empirical specifications

3.1 Investment model

At date 1, firm i has a growth opportunity with date-2 payoff

$$G(K_i, \tilde{\theta}_i) = \tilde{\theta}_i K_i - \frac{K_i^2}{2}, \quad (4)$$

where K_i is investment and $\tilde{\theta}_i$ is the marginal productivity of investment.

The manager chooses investment to maximize expected payoff conditional on information Ω_1 :

$$\max_{K_i} E \left[G(K_i, \tilde{\theta}_i) \mid \Omega_1 \right]. \quad (5)$$

The first-order condition implies

$$K_i^*(\Omega_1) = E(\tilde{\theta}_i \mid \Omega_1). \quad (6)$$

Interpretation: Investment equals the manager's expectation of the marginal return to investment. Any signal that changes beliefs can change investment.

3.2 Signals

The manager observes internal and external signals:

$$s_{m,i} = \tilde{\theta}_i + \chi_i, \quad (7)$$

$$P_i = \tilde{\theta}_i + u_i, \quad (8)$$

$$P_{-i} = \tilde{\theta}_{-i} + u_{-i}, \quad (9)$$

$$\tilde{\theta}_{-i} = \rho_i \tilde{\theta}_i, \quad (10)$$

$$s_{u,i} = u_i + \eta_i, \quad (11)$$

$$s_{u,-i} = u_{-i} + \eta_{-i}. \quad (12)$$

Here P_i is the firm's own stock-price signal, P_{-i} is the peers' stock-price signal, and u_i, u_{-i} are noise components. The error variances in η_i and η_{-i} determine how well managers can filter out price noise.

Interpretation: The faulty informant hypothesis is that managers use prices but cannot perfectly identify the noise in them.

3.3 Projection form of the optimal investment rule

The key theoretical implication is that optimal investment can be written as

$$K_i^* = \gamma_i P_i^* + \alpha_i u_i^o + \gamma_{-i} P_{-i}^* + \alpha_{-i} u_{-i}^o + \varepsilon_i, \quad (13)$$

where P_i^* and P_{-i}^* are residual price components, u_i^o and u_{-i}^o are observable components of price noise, and ε_i is orthogonal to the regressors.

The model predicts:

$$\alpha_i > 0, \quad |\alpha_{-i}| \neq 0 \quad (14)$$

when managers cannot fully filter out the noise in stock prices.

Interpretation: The coefficient on peer-price noise is the clean test. Financing and agency channels linked to the firm's own price do not generally predict that noise in peers' prices should affect the focal firm's investment after controlling for the firm's own price.

3.4 Why price sensitivity is not enough

Even if managers perfectly filter out noise, investment can still be sensitive to stock prices because stock prices convey fundamental information. Therefore, regressing investment on peer prices does not test the faulty informant channel.

Interpretation: The paper's key contribution is to estimate investment sensitivity to *noise in stock prices*, not merely sensitivity to stock prices.

3.5 Baseline empirical specification

The baseline OLS specification is

$$I_{i,t} = \lambda_i + \delta_t + \alpha_0 \text{MFHS}_{-i,t-1} + \gamma_0 Q_{-i,t-1}^* + \alpha_1 \text{MFHS}_{i,t-1} + \gamma_1 Q_{i,t-1}^* \quad (15)$$

$$+ \Gamma X_{i,-i,t-1} + \epsilon_{i,t}. \quad (16)$$

- $I_{i,t}$ is Capex/ PPE .
- λ_i are firm fixed effects.
- δ_t are year fixed effects.
- $\text{MFHS}_{-i,t-1}$ is peer fire-sale price pressure.
- $Q_{-i,t-1}^*$ is the residual component of peers' stock prices.
- $X_{i,-i,t-1}$ contains firm and peer controls.

Interpretation: The coefficient of interest is α_0 . If managers fully recognize peer-price noise, $\alpha_0 = 0$. If they mistake the noise for information, $\alpha_0 \neq 0$.

3.6 Division-level specification

For conglomerates, the paper decomposes the stock price of division peers using

$$Q_{-i,d,t} = \lambda_{i,d} + \delta_{i,t} + \phi \text{MFHS}_{-i,d,t} + \hat{\nu}_{-i,d,t}, \quad (17)$$

then estimates

$$I_{i,d,t} = \lambda_{i,d} + \delta_{i,t} + \alpha_0 \text{MFHS}_{-i,d,t-1} + \gamma_0 Q_{-i,d,t-1}^* + \Gamma X_{-i,d,t-1} + \epsilon_{i,d,t}. \quad (18)$$

Interpretation: Firm-year fixed effects absorb all firm-level time-varying omitted variables. The estimate is identified from capital allocation across divisions of the same firm in the same year.

4 Identification and empirical design

4.1 What identifies the faulty-informant effect

- Mutual-fund outflows create hypothetical selling pressure that moves stock prices temporarily.
- The paper uses the price pressure on *peers*, not only on the firm itself.
- The baseline controls for the focal firm's own stock-price components.
- Firm fixed effects remove time-invariant firm heterogeneity.
- Year fixed effects remove aggregate investment and market conditions.

Interpretation: A positive coefficient on $MFHS_{-i}$ means that when peers' prices are pushed down by nonfundamental selling pressure, the focal firm reduces investment.

4.2 Evidence that MFHS is nonfundamental

- Figure 1 shows stocks exposed to fire-sale pressure experience abnormal negative returns that subsequently reverse.
- Figure 2 shows that the fire-sale pressure is not concentrated in one year or industry.
- Figure 3 shows insiders buy after the stock-price drop, consistent with insiders viewing the drop as nonfundamental.

Interpretation: These tests support using MFHS as an ex post observable component of price noise.

4.3 Why peer-price noise is cleaner than own-price noise

- Own-price noise may affect investment through equity issuance, managerial compensation, takeover threats, or catering.
- These mechanisms operate through the firm's own stock price.
- They do not naturally imply that noise in product-market peers' stock prices affects the focal firm's investment after controlling for its own price.

Interpretation: The peer design isolates the information-learning channel more cleanly than own-price feedback tests.

4.4 Main credibility checks

- Multiple peer aggregation methods produce similar results.
- Lead-lag tests show the noise effect is temporary, while fundamental price information is more persistent.
- Cross-sectional tests show stronger effects where managers are less able to filter price noise and more likely to learn from peer prices.
- Division-level regressions with firm-year fixed effects control for firm-level omitted variables.
- Alternative channels related to financing, CEO pressure, peer investment mimicking, and correlated own-price noise do not explain the result.

Interpretation: The paper builds identification by showing the same mechanism in timing, cross section, and within-firm allocation tests.

4.5 What the paper cannot fully prove

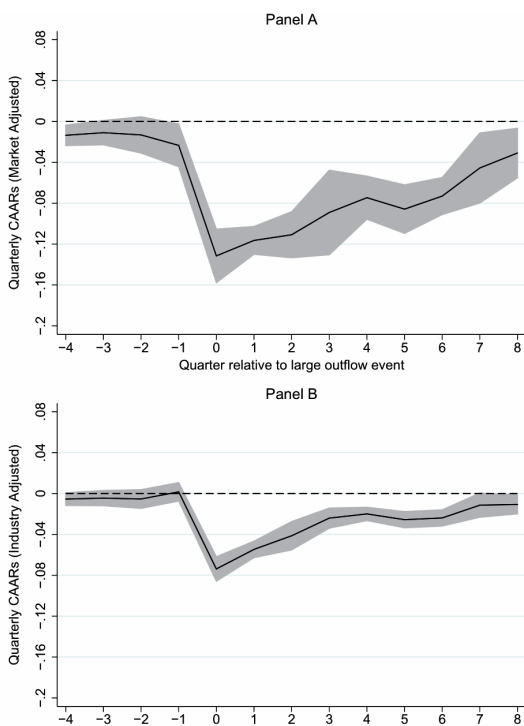
- It cannot directly observe managers' beliefs.
- It observes only one measurable component of price noise.
- The residual price component Q^* is not guaranteed to be purely fundamental.
- The identification assumes hypothetical mutual-fund sales are not proxying for omitted fundamentals of the focal firm through correlated peer shocks.

Interpretation: These limits are why the robustness and alternative-channel tests are central to the paper.

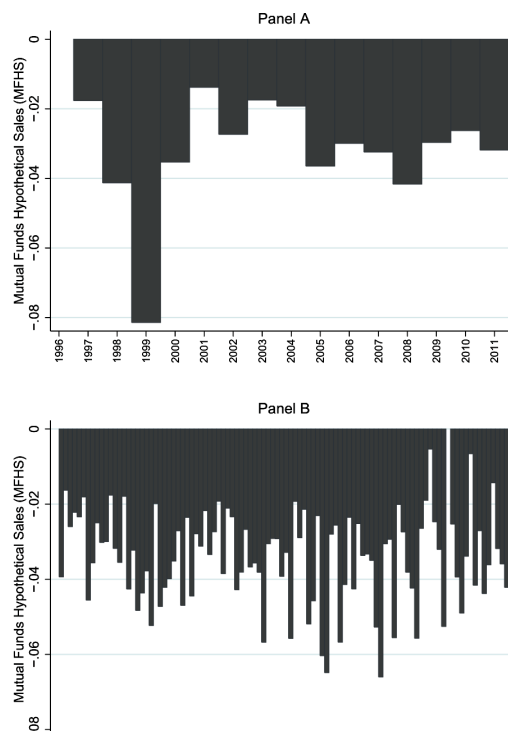
5 Tables and figures

The original figures and tables are directly cropped from the paper and grouped to save space.

5.1 Figures 1–2: Validating the mutual-fund fire-sale measure



(a) Effect of mutual fund hypothetical sales on stock prices



(b) MFHS across time and industries

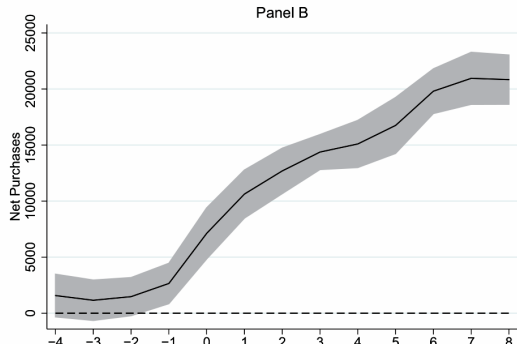
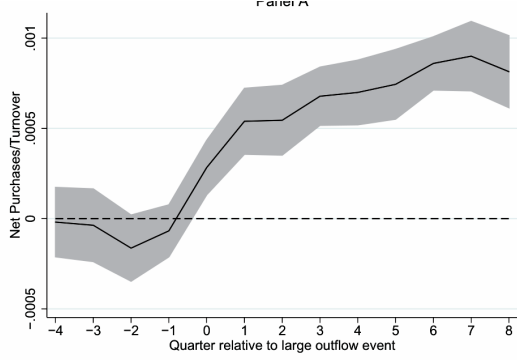
Read:

- Figure 1 shows a sharp stock-price decline after large mutual-fund outflow events.
- The decline reverses over roughly two years, which is consistent with temporary price pressure rather than permanent fundamental deterioration.

- Figure 2 shows MFHS varies across years and industries, with no single time or industry completely driving the measure.

Economic message: The fire-sale instrument has the two properties the paper needs: it moves prices and the movement looks nonfundamental.

5.2 Figure 3 and Table 1: Insider validation and sample summary



(a) Insiders' net purchases around fire-sale events

Table 1
Summary statistics

Variable	Mean	SD	Min.	Max.	Obs.
$MFHS_i$	-0.033	0.055	-0.538	0.000	45,275
Q_i	1.958	1.473	0.547	10.010	45,275
CF/A_i	0.018	0.216	-1.167	0.361	45,275
$Size_i$	5.622	1.929	1.290	10.644	45,275
$Capex/PPE_i$	0.349	0.388	0.008	2.524	45,275
$MFHS_{-i}$	-0.031	0.029	-0.485	0.000	45,275
$\bar{Q}_{-i}$	2.073	0.848	0.547	10.010	45,275
CF/A_{-i}	0.013	0.109	-1.167	0.361	45,275
$Size_{-i}$	5.759	1.060	1.290	10.644	45,275
$Capex/PPE_{-i}$	0.380	0.221	0.008	2.524	45,275

This table reports summary statistics for the main employed variables. For each variable, we present its mean, minimum, and maximum; its standard deviation; and the number of nonmissing observations. Table B1 defines all variables. Statistics for a firm are indexed by i and statistics for peers' average (i.e., the average of peers for each firm-year observation) are indexed by $-i$. Averages are computed by excluding firm i itself. Peers are defined by the TNIC industries developed by Hoberg and Phillips (2016). The sample period is from 1996 to 2011. All variables are winsorized at the 1% level in each tail.

(b) Summary statistics

Read:

- Figure 3 shows insiders buy after fire-sale shocks, both in turnover-scaled and share-count terms.
- Insider trading against the price pressure supports the view that the price movements are not fully fundamental.
- Table 1 shows 45,275 observations, average $Capex/PPE$ of 0.349, average firm Q of 1.958, and average peer Q of 2.073.

Economic message: The summary statistics establish the baseline economic scale, and insider purchases provide an additional validation of the nonfundamental shock.

5.3 Table 2: Main investment-to-noise result

Table 2
Main results: Investment-to-noise sensitivity

Dependent variable: $Capex/PPE_i$

Peers' aggregation:	E-W (1)	S-W (2)	Median (3)	5 closest (4)	Agg. (5)
$MFHS_{-i}$	0.015*** (5.50)	0.014*** (5.69)	0.009*** (4.42)	0.014*** (5.15)	0.011* (2.03)
$\bar{Q}_{-i}$	0.024*** (4.96)	0.025*** (5.17)	0.025*** (3.83)	0.019*** (4.96)	0.015*** (3.66)
$\bar{CF}/A_{-i}$	0.019*** (3.39)	0.011* (2.08)	0.015*** (3.29)	0.011** (2.11)	0.005 (1.38)
$\bar{Size}_{-i}$	0.002 (0.40)	0.001 (0.22)	-0.001 (-0.21)	-0.000 (-0.03)	0.001 (0.14)
$Capex/PPE_{-i}$	0.028*** (3.85)	0.014** (2.28)	0.027*** (3.77)	0.023*** (4.66)	0.016** (2.70)
$MFHS_i$	0.011*** (5.92)	0.011*** (6.02)	0.011*** (6.47)	0.012*** (6.30)	0.012*** (5.52)
Q_i^*	0.080*** (11.78)	0.080*** (12.03)	0.081*** (12.51)	0.081*** (11.73)	0.085*** (11.36)
CF/A_i	0.037*** (12.11)	0.036*** (12.24)	0.037*** (12.75)	0.037*** (11.65)	0.038*** (11.36)
$Size_i$	-0.080*** (-2.91)	-0.078** (-2.80)	-0.077** (-2.80)	-0.077*** (-2.82)	-0.073** (-2.52)
Obs.	45,275	45,275	45,275	45,275	45,275
Firm FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.371	0.370	0.371	0.370	0.367

This table presents OLS estimations of specification (9). The dependent variable is the investment of firm i in year t , defined as capital expenditures, divided by lagged property, plant, and equipment (PPE). $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows ("price pressure") of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i}$ is the error term v_{-i}^* estimated from specification (8) and corresponds to the component of peers' stock price that is unexplained by mutual fund hypothetical sales. The subscript $-i$ for a variable refers to a portfolio that aggregates the peers of firm i . In Column 1, we use equally weighted averages. In Column 2, we use weighted averages, where weights are product description similarity scores from Hoberg and Phillips (2015). In Column 3, we use medians. In Column 4, we use equally weighted averages computed across the five "closest" peers, using similarity score as a distance measure. In Column 5, we aggregate all variables across firm i 's peers before computing ratios. All explanatory variables are divided by their sample standard deviation to facilitate economic interpretation. Table B1 defines all variables. Standard errors used to compute t -statistics (in parentheses) are clustered in two ways: by industry

Figure 3: Main results: Investment-to-noise sensitivity

Read:

- Column 1 estimates $\alpha_0 = 0.015$ with a t -statistic of 5.50.
- Since MFHS is negative when fire-sale pressure is strong, a one-standard-deviation decrease in peers' noise component lowers investment by about 1.5 percentage points.
- Peer residual stock-price information Q_{-i}^* is also positive and significant, with coefficient 0.024 in Column 1.
- The $MFHS_{-i}$ coefficient remains positive across equal-weighted, similarity-weighted, median, five-closest, and aggregate peer definitions.

Economic message: Managers respond to the nonfundamental component of peers' prices, not only to information in prices.

5.4 Figure 4 and Table 3: Dynamic response and opportunity cost

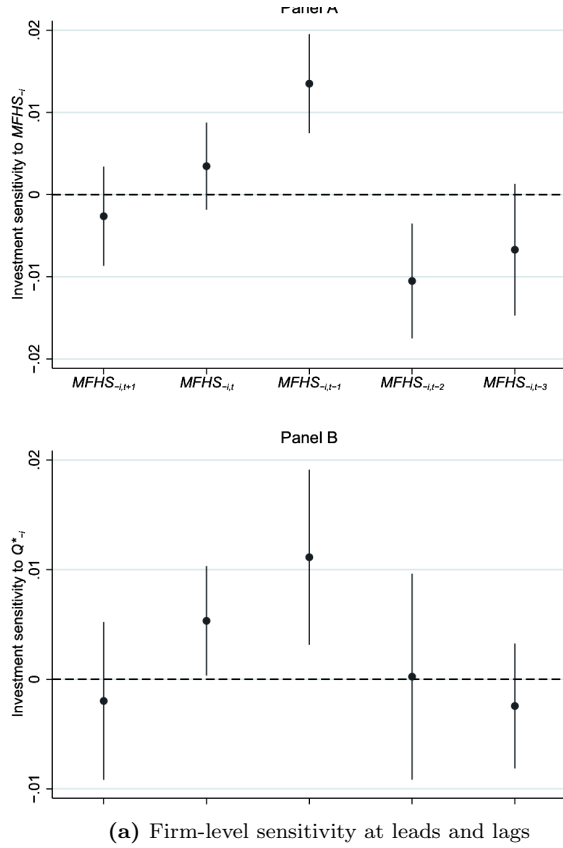


Table 3
Opportunity costs estimates

		Average unit NPV					
		0.35	0.45	0.55	0.65	0.75	0.85
Discount rate	10.0%	14.0	18.0	12.0	26.0	30.0	34.0
	12.5%	17.1	22.0	26.9	31.8	36.7	41.6
	15.0%	20.1	25.8	31.6	37.3	43.0	48.8

This table reports the results of simulating the total opportunity cost associated with the faulty informant channel over the period 1996–2011. We simulate the value impact for shareholders of postponing \$440 billion of capital expenditure for a year assuming that the average NPV of \$1 dollar invested ranges between \$0.35 and \$0.96 and that the appropriate discount rate ranges between 10% and 15%.

(b) Opportunity cost estimates

Read:

- Figure 4 shows the peer-noise effect is temporary: investment falls after noise but the effect reverses.
- Fundamental peer-price information has a more persistent effect.
- Table 3 converts transient investment losses into shareholder opportunity-cost estimates.
- Depending on unit NPV and discount-rate assumptions, the total opportunity cost ranges from about \$14 billion to \$55.1 billion over the sample, or roughly \$0.9–\$3.7 billion per year.

Economic message: Noise causes firms to postpone or miss positive-NPV investment opportunities, producing economically large but transient real effects.

5.5 Table 4: Cross-sectional mechanisms

Table 4
Cross-sectional results

Dependent variable: $Capex/PPE_i$

Cross-sectional variation of interest: Information about Noise $\sigma_{\eta,-i}^2$

Proxy variable:

	Information about Noise $\sigma_{\eta,-i}^2$		correlation of fundamentals $ \rho_i $		Peers' stock price informativeness $\sigma_{\eta,-i}^2$		Information about fundamentals $\sigma_{\eta,-i}^2$	
	<i>Common Ownership</i> $_{i,-i}$ (1)	<i>Analyst</i> $_{i,-i}$ (2)	<i>Sales</i> $_{i,-i}$ (3)	<i>Assets</i> $_{i,-i}$ (4)	<i>BPS</i> $_{i,-i}$ (5)	<i>AnalystFE</i> $_{i,-i}$ (6)	<i>Insider</i> $_{i,-i}$ (7)	<i>Market</i> $_{i,-i}$ (8)
$MFHS_{-i}$	0.023*** (5.09)	0.021*** (5.12)	0.007 (1.67)	0.008** (2.12)	0.014*** (5.81)	0.021*** (4.53)	0.015*** (5.40)	0.017*** (4.82)
$MFHS_{-i} \times \sigma_{\eta,-i}^2$	-0.006*** (-2.79)	-0.006** (-2.61)						
$MFHS_{-i} \times \rho_i $			0.007*** (3.08)	0.004* (1.82)				
$MFHS_{-i} \times \sigma_{\eta,-i}^2$					0.002* (1.78)	0.006*** (2.84)		
$MFHS_{-i} \times \sigma_{\eta,-i}^2$							-0.002 (-0.90)	-0.004* (-1.94)
Obs.	45,275	33,350	44,137	42,751	45,275	44,408	45,275	45,127
Controls (interacted)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	.376	.379	.379	.353	.372	.375	.372	.372

This table shows how investment-to-noise sensitivity varies in the cross-section of firms along four dimensions: the quality of managerial information about noise in peers' stock price ($\sigma_{\eta,-i}^2$), the degree of correlation between the fundamentals of the firm and the fundamentals of its peers (ρ_i) in absolute value, the informativeness of peers' stock price ($\sigma_{\eta,-i}^2$), and the precision of managerial information about firm fundamentals ($\sigma_{\eta,-i}^2$). The table presents OLS estimations of specification (9), where all explanatory variables are interacted with proxies for the cross-sectional variation of interest. The dependent variable is the investment of firm i in year t , defined as capital expenditures divided by lagged property, plant, and equipment (PPE). In Column 1, the proxy for $\sigma_{\eta,-i}^2$ is an index of mutual funds ownership overlap between firm i and its peers. In Column 2, the proxy for $\sigma_{\eta,-i}^2$ is the average difference between analyst target price and current stock price for every peer of firm i . In Column 3, the proxy for ρ_i is the average pairwise correlation of the (log of) sales. In Column 4, the proxy for ρ_i is the average pairwise correlation of the (log of) assets. In Column 5, the proxy for $\sigma_{\eta,-i}^2$ is the measure of price informativeness proposed by Bai, Philippon, and Savov (2016). In Column 6, the proxy for $\sigma_{\eta,-i}^2$ is the average earnings forecast error by financial analysts covering peers' stock prices. In Column 7, the proxy for $\sigma_{\eta,-i}^2$ is the profitability of insiders' trades for firm i . In Column 8, the proxy for $\sigma_{\eta,-i}^2$ is the market share of firm i . $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows ("price pressure") of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . Table B1 defines all variables. All explanatory variables are interacted with the proxy variable, and this proxy variable is included as a control in all specifications. All explanatory variables are divided by their sample standard deviation to facilitate economic interpretation. Standard errors used to compute t -statistics (in parentheses) are clustered in two ways: by industry (FIC300) and by year. * $p < .01$; ** $p < .05$; *** $p < .001$.

Figure 5: Cross-sectional results

Read:

- The investment-to-noise sensitivity varies with managers' ability to identify noise and with the information content of peer prices.
- The effect is stronger when mutual-fund ownership overlap or analyst information about peers is lower.
- It is also stronger when peers' fundamentals are more correlated with the focal firm's fundamentals.
- The patterns line up with the faulty informant story: managers respond more to noisy peer prices when those prices are useful signals but the noise is harder to filter.

Economic message: The cross section strengthens the learning interpretation. The effect is not uniform; it appears where the theory says noisy prices should matter most.

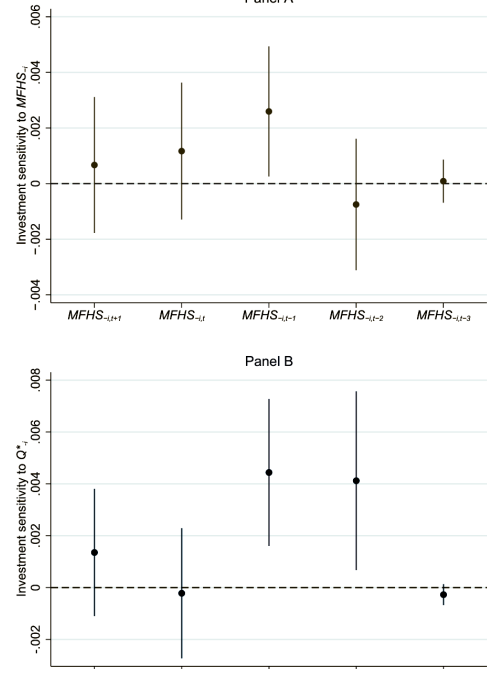
5.6 Table 5 and Figure 5: Within-conglomerate capital allocation

Table 5
Within-conglomerate investment

Dependent variable:	$Capex/Assets_{i,t}$					
Industry:	FF49 (1)	SIC2 (2)	NAICS3 (3)	FF49 (4)	SIC2 (5)	NAICS3 (6)
$MFHS_{-i,t}$	0.003* (1.93)	0.003*** (2.64)	0.003** (2.57)	0.003* (1.75)	0.003** (2.55)	0.004** (2.62)
$\bar{Q}_{-i,t}^*$	0.008*** (5.05)	0.007*** (3.39)	0.003* (1.71)	0.007*** (5.25)	0.007*** (3.39)	0.004* (1.96)
$\bar{CF}/A_{-i,t}$	0.004*** (2.70)	0.001 (0.97)	0.003 (1.57)	0.003** (2.58)	0.002 (1.31)	0.003 (1.49)
$\bar{Size}_{-i,t}$	0.004 (1.08)	0.008* (2.00)	0.005 (1.24)	0.003 (1.01)	0.007* (1.77)	0.003 (0.76)
$\bar{Capex}/\bar{PPE}_{-i,t}$	0.005*** (3.26)	0.005** (2.55)	0.004*** (2.67)	0.005*** (3.05)	0.005** (2.59)	0.003** (2.47)
Obs.	62,822	63,263	40,876	62,818	62,242	40,809
Firm-year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Firm-division FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	.369	.365	.396	.369	.367	.396

This table presents OLS estimations of specification (11). The dependent variable is the investment of division d of firm i in year t , defined as capital expenditures divided by lagged total assets. $MFHS_{-i,t}$ is the average hypothetical stock sales due to mutual funds large outflows ("price pressure") of all firms operating in the same industry as division d of firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i,t}^*$ is the error term $v_{-i,t}$ estimated from Equation (10) and corresponds to the component of division peers' stock price unexplained by mutual funds hypothetical sales. In Column 1, we define industry using the Fama-French 49 classification (FF49), in Column 2 we define industry using the two-digit Standard Industry Classification (SIC2), and, in Column 3, we define industry using the three-digit North American Industry Classification System (NAICS3). In Columns 4 to 6, we perform the same tests used in Columns 1 to 3, except that we restrict the definition of peers to single-division firms from the same industry. Table B1 defines all variables. The subscript $-i$ for a variable refers to the (equally weighted) average value of the variable across peers of division d of firm i . All explanatory variables are divided by their sample standard deviation to facilitate economic interpretation. Standard errors used to compute t -statistics (in parentheses) are clustered in two ways: by industry (FF49 in Columns 1 and 4, SIC2 in Columns 2 and 5, and NAICS3 in Columns 3 and 6) and by year. * $p < .01$; ** $p < .05$; *** $p < .001$.

(a) Within-conglomerate investment



(b) Division-level sensitivities

Read:

- Table 5 estimates the same mechanism at the division level for conglomerates.
- With firm-year fixed effects, the coefficients on division-peer MFHS are positive and significant across FF49, SIC2, and NAICS3 industry definitions.
- Figure 5 shows division-level dynamic patterns similar to the firm-level tests.

Economic message: The result survives a very demanding control: comparing divisions of the same firm in the same year. This makes firm-level omitted variables much less plausible.

5.7 Tables 6–7: Financing and pressure channels

Table 6

Alternative explanation: Financing channel

Dependent variable:	CDS spread _i (1)	Debt spread _i (2)	Debt -Cons _i (3)	Equity -Cons _i (4)	Payout _i (5)	Security issue _i (6)	Capex /PPE _i (7)
$MFHS_{-i}$	0.077 (0.93)	0.035*** (2.97)	-0.000 (-0.22)	0.001 (1.02)	-0.001 (-1.32)	0.004 (1.59)	0.014*** (4.85)
$\bar{Q}_{-i}$	0.027 (0.38)	-0.022 (-1.41)	-0.001 (-1.56)	0.000 (0.58)	-0.001** (-2.29)	0.002 (1.17)	0.023*** (4.97)
$\overline{CF/A}_{-i}$	-0.412*** (-3.81)	-0.061** (-2.42)	-0.001 (-0.59)	-0.001 (-1.08)	0.001 (1.63)	0.006* (2.02)	0.019*** (3.41)
$\overline{Size}_{-i}$	-0.099 (-1.13)	-0.009 (-0.39)	0.000 (0.57)	0.002 (1.42)	0.000 (0.15)	-0.001 (-0.32)	0.002 (0.47)
$\overline{Capex/PPE}_{-i}$	-0.015 (-0.10)	-0.013 (-0.61)	0.002* (1.98)	-0.000 (-0.17)	-0.000 (-0.51)	-0.005 (-1.49)	0.029*** (3.83)
$MFHS_i$	-0.350 (-1.26)	0.005 (0.35)	-0.000 (-0.32)	0.001 (0.90)	-0.001** (-2.29)	0.002 (1.42)	0.009*** (5.12)
Q_i^*	-0.115* (-1.92)	-0.130*** (-11.18)	-0.001* (-1.76)	0.002*** (6.17)	0.002** (2.57)	0.016*** (5.90)	0.079*** (11.33)
CF/A_i	-1.141*** (-3.62)	-0.358*** (-6.08)	-0.001** (-2.41)	-0.006*** (-9.68)	0.005*** (5.78)	-0.032*** (-8.66)	0.041*** (13.66)
$Size_i$	-0.887 (-1.65)	-0.589*** (-10.73)	-0.000 (-0.09)	-0.001 (-0.38)	0.015*** (5.45)	-0.106*** (-7.77)	-0.066*** (-2.45)
$Payout/A_i$							-0.012*** (-3.69)
$SecurityIssue/A_i$							0.024*** (3.87)
Obs.	3,763	10,734	33,102	33,102	45,275	43,347	43,347
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	.650	.640	.484	.591	.359	.463	.370

This table presents OLS estimations of specification (9), where we use firm-level measures of financing costs and access to external capital as dependent or control variables. $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows ("price pressure") of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i}$ is the error term v_{-i} estimated from specification (8) and corresponds to the component of peers' stock price that is unexplained by mutual fund hypothetical sales. Dependent variables include the credit default swap (CDS) spread of firm i in year t (Column 1); the average spread of firm i in year

(a) Alternative explanation: financing channel

Table 7

Alternative explanation: Pressure channel

Dependent variable:	HHI _i (1)	Nb.Peers _i (2)	Capex/PPE _i (3)	Capex/PPE _i RPE=1 (4)	Capex/PPE _i RPE=0 (5)	Prob(Target) _i (6)	CEO turnover _i (7)
$MFHS_{-i}$	0.002 (0.76)	-0.000 (-0.02)	0.015*** (5.59)	0.016*** (3.56)	0.015** (2.69)	0.004 (1.13)	0.002 (0.57)
$\bar{Q}_{-i}$	0.000 (0.32)	0.020*** (2.81)	0.023*** (4.95)	0.026*** (4.74)	0.019*** (4.27)	-0.005** (-2.52)	0.005 (1.22)
$\overline{CF/A}_{-i}$	0.003 (0.98)	-0.063*** (-3.96)	0.019*** (3.40)	0.025*** (4.14)	0.009 (1.23)	0.004* (1.76)	-0.000 (-0.11)
$\overline{Size}_{-i}$	-0.010*** (-2.84)	0.015 (1.10)	0.002 (0.43)	-0.005 (-0.85)	0.009 (1.18)	0.002 (0.40)	0.007 (1.58)
$\overline{Capex/PPE}_{-i}$	-0.009*** (-5.46)	0.037*** (3.43)	0.028*** (3.86)	0.020** (2.17)	0.037*** (4.75)	-0.005 (-1.66)	-0.004 (-0.73)
$MFHS_i$	0.002 (1.08)	-0.014*** (-2.25)	0.011*** (5.86)	0.010*** (3.45)	0.011*** (5.53)	-0.007** (-2.74)	-0.003 (-1.08)
Q_i^*	-0.001 (-0.87)	0.005 (1.21)	0.080*** (11.76)	0.081*** (10.93)	0.077*** (9.36)	-0.009*** (-3.20)	-0.010** (-2.29)
CF/A_i	-0.005*** (-3.46)	0.028*** (4.15)	0.037*** (12.15)	0.031*** (5.68)	0.041*** (6.09)	-0.015*** (-3.94)	-0.036*** (-3.72)
$Size_i$	-0.033*** (-6.36)	0.179*** (7.80)	-0.080*** (-2.91)	-0.073** (-2.73)	-0.089*** (-3.34)	0.061*** (4.72)	0.006 (0.34)
$Prob(Acquirer)_{-i}$			0.001 (0.39)				
$Prob(Target)_{-i}$			-0.005** (-2.42)				
Obs.	45,275	45,275	45,275	23,455	21,820	45,275	18,107
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	.582	.855	.371	.395	.352	.163	.010

This table presents OLS estimations of specification (9) where we use measures of pressure on CEO's incentives to invest as independent or control variables. $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows ("price pressure") of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i}$ is the error term v_{-i} estimated from specification (8) and corresponds to the component of peers' stock price that is unexplained by mutual fund hypothetical sales. The dependent variable is the Herfindahl-Hirschman index of sales computed over all years of firm i in year t in Column 1 and the number of years of

(b) Alternative explanation: pressure channel

Read:

- Table 6 shows peer-price noise is not mainly operating through financing costs, debt constraints, equity constraints, payout, or security issuance.
- The investment result remains when these financing-related measures are used or controlled for.
- Table 7 tests pressure channels linked to competition, relative performance evaluation, takeover pressure, and CEO turnover.
- The evidence does not support these channels as the explanation for the baseline investment response.

Economic message: The effect is not simply that noisy peer prices tighten financing or force managers to mimic competitors. It is more consistent with a belief-updating channel.

5.8 Tables 8–9: Investment mimicking and correlated noise

Table 8
Alternative explanation: Investment-mimicking channel

Dependent variable:	$Capex/PPE_i$		
	(1)	(2)	(3)
$MFHS_{-i}$	0.024** (2.20)	0.016*** (5.49)	0.007*** (3.28)
$\bar{Q}_{-i}^*$	0.033** (2.63)	0.028*** (4.90)	0.014*** (4.15)
$\overline{CF/A}_{-i}$	0.008 (0.50)	0.002 (0.20)	0.015*** (3.30)
$\overline{Size}_{-i}$	0.005 (0.35)	0.011** (2.11)	0.002 (0.43)
$MFHS_i$	0.012** (2.71)	0.010*** (3.32)	0.010*** (5.65)
Q_i^*	0.075*** (7.50)	0.080*** (7.57)	0.078*** (11.88)
CF/A_i	0.032** (2.27)	0.041*** (5.61)	0.036*** (11.72)
$Size_i$	-0.098*** (-4.33)	-0.097*** (-3.66)	-0.080*** (-3.01)
$Capex/PPE_{-i}$			0.024*** (3.85)
$Contemporaneous Capex/PPE_{-i}$			0.047*** (5.80)
Obs.	9,124	21,046	45,244
Firm FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Ind-year FE	—	—	—
Adj. R^2	.396	.389	.375

This table presents OLS estimations of specifications similar to specification (9). The dependent variable is the investment of firm i in year t , defined as capital expenditures divided by lagged property, plant, and equipment (PPE). $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows (“price pressure”) of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i}^*$ is the error term v_{-i}^* estimated from specification (8) and corresponds to the component of peers’ stock price that is unexplained by mutual fund hypothetical sales. The subscript $-i$ for a variable refers to a portfolio that aggregates the noise

(a) Alternative explanation: investment-mimicking channel

Table 9
Alternative explanation: Correlated noise

Dependent variable:	$Capex/PPE$			
	(1)	(2)	(3)	(4)
$MFHS_{-i}$	0.015*** (5.50)		0.015*** (5.33)	
$\bar{Q}_{-i}^*$	0.024*** (4.96)	0.023*** (4.46)	0.024*** (4.97)	0.023*** (4.44)
$\overline{CF/A}_{-i}$	0.019*** (3.39)	0.018*** (3.29)	0.019*** (3.39)	0.018*** (3.28)
$\overline{Size}_{-i}$	0.002 (0.40)	0.001 (0.29)	0.002 (0.40)	0.001 (0.29)
$Capex/PPE_{-i}$	0.028*** (3.85)	0.030*** (4.11)	0.028*** (3.86)	0.030*** (4.13)
$MFHS_i$	0.011*** (5.92)	0.013*** (6.14)		
Q_i^*	0.080*** (11.78)	0.082*** (11.52)		
CF/A_i	0.037*** (12.11)	0.036*** (12.17)	0.037*** (12.16)	0.036*** (12.25)
$Size_i$	-0.080*** (-2.91)	-0.079*** (-2.88)	-0.080*** (-2.91)	-0.079*** (-2.89)
$\bar{Q}_i$			0.122*** (11.79)	0.124*** (11.48)
Obs.	45,275	45,275	45,275	45,275
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Adj. R^2	.371	.371	.371	.371

This table presents the results from estimations of specifications similar to that of Equation (9). The dependent variable is the investment of firm i in year t , defined as capital expenditures divided by lagged property, plant, and equipment (PPE). $MFHS_{-i}$ is the average hypothetical stock sales due to mutual funds large outflows (“price pressure”) of all firms belonging to the same TNIC industry as firm i in year $t-1$, excluding firm i . $\bar{Q}_{-i}^*$ is the error term v_{-i}^* estimated from specification (8) and corresponds to the component of peers’ stock price that is unexplained by mutual fund hypothetical sales. The subscript $-i$ for a variable refers to an equally

(b) Alternative explanation: correlated noise

Read:

- Table 8 shows the effect remains when peers do not change investment, when peers increase investment, and when contemporaneous peer investment is controlled for.
- Table 9 shows the effect is robust to specifications designed to address correlated noise between the focal firm’s stock price and peers’ stock prices.
- The coefficient on $MFHS_{-i}$ remains close to the baseline value of 0.015 in the relevant specifications.

Economic message: These tests rule out the possibility that firms merely mimic peers’ investment or that peer-price noise is just proxying for the firm’s own price noise.

6 Short summary

- The paper studies whether nonfundamental stock-price movements affect real investment.
- The key channel is that managers use stock prices as signals but cannot perfectly distinguish information from noise.
- The test uses noise in product-market peers’ stock prices rather than the firm’s own stock price.
- Nonfundamental noise is measured using mutual-fund hypothetical sales caused by large investor outflows.
- Stocks exposed to fire-sale pressure experience temporary price drops that later reverse.
- Insiders buy after these shocks, supporting the idea that the shocks are nonfundamental.
- The baseline coefficient on peer-price noise is 0.015, implying a 1.5 percentage point decrease in investment after a one-standard-deviation negative shock.
- This is about 4.3% of average investment.
- The estimated aggregate investment loss is about \$29 billion per year.

- Opportunity costs for shareholders are estimated at roughly \$0.9–\$3.7 billion per year.
- The noise effect is temporary, while the effect of fundamental peer-price information is more persistent.
- The effect is stronger where managers are less able to filter noise or more likely to learn from peer prices.
- Division-level tests with firm-year fixed effects show that noise affects capital allocation within conglomerates.
- Financing, managerial pressure, investment mimicking, and correlated-own-noise explanations do not overturn the result.

7 Conclusion

7.1 Core conclusion

Dessaint, Foucault, Frésard, and Matray show that stock-market noise can affect corporate investment through a learning channel. Managers observe prices because prices contain information, but if managers cannot perfectly filter the noise in those prices, nonfundamental price movements can distort real investment decisions.

7.2 Economic intuition

A manager deciding how much to invest wants to infer the profitability of a growth opportunity. Product-market peers' stock prices are useful signals because peer fundamentals are correlated with the firm's fundamentals. But if peer prices fall because of mutual-fund fire sales, the manager may partially interpret that decline as bad news about investment opportunities. The result is underinvestment even though the price decline is nonfundamental.

7.3 Takeaway

The paper broadens the real-effects literature. Stock-market inefficiency can affect firms even without equity issuance, financing constraints, short-termism, or agency problems. The real cost comes from information aggregation: prices inform managers, but noisy prices sometimes misinform them.